

1(a) Describe the primary structure of a collagen polypeptide. [2]

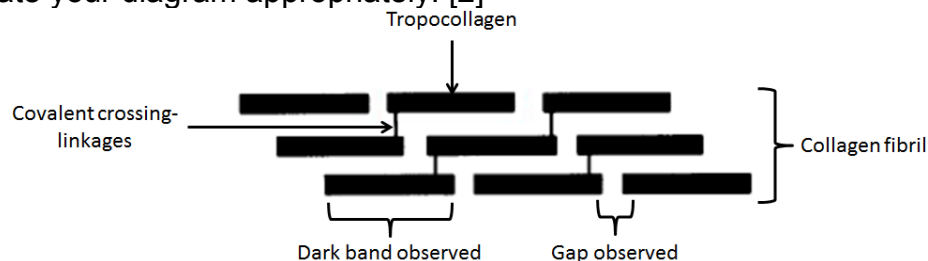
- Primary structure is the number, type and sequence of amino acids [1/2]
- joined together by peptide bonds [1/2]
- The primary structure of collagen is rich in amino acid proline and glycine [1/2]
- Also contains modified amino acid hydroxyproline and hydroxylysine [1/2]
- (repetitive sequence) Every third amino acid in the polypeptide sequence is glycine [1/2]

(b) With reference to **Fig. 1.1**,

(i) describe two ways how the structure of a collagen fibre differs from that of an enzyme such as amylase. [2]

- A collagen fibre forms a strand-like / linear structure [1/2] while amylase has a globular / spherical shape [1/2]
- A collagen fibre is made up of the aggregation of many polypeptide chains cross-linked together [1/2] while amylase is made up of only one polypeptide chain [1/2]
- A collagen fibre has a quaternary structure [1/2] while amylase has a tertiary structure [1/2]

(ii) Draw a diagram to show why a collagen fibril appears as a banded structure. Annotate your diagram appropriately. [2]



- Staggered arrangement of tropocollagen [1/2]
- Formation of covalent cross-linkages between tropocollagen [1/2]
- Reasons for banded appearance of collagen fibril [1]
- Max 1.5 marks if banded appearance not clearly shown/explained

(c) **Fig. 1.2** shows two membrane-bound organelles commonly found in animal cells.

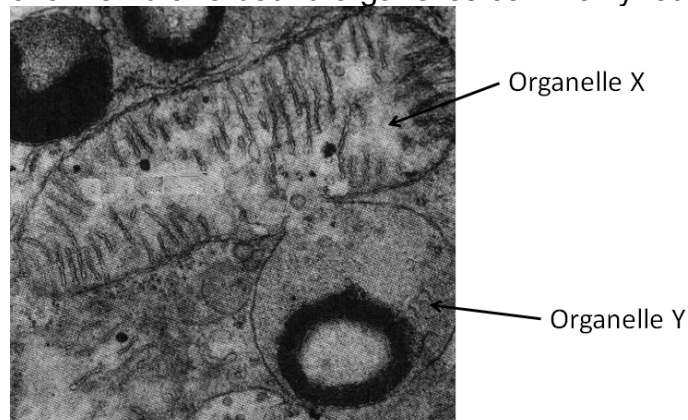


Fig 1.2

(i) State the name of the process occurring in **Fig. 1.2**.

- Autophagy [1]

(ii) Describe two other functions of organelle Y not shown in Fig. 1.2. [2]

- Organelle Y: Lysosome [1/2]
- Fuse with endocytotic vesicles [1/2] to digest the materials within [1/2]
- Release hydrolytic enzymes [1/2] resulting in self-digestion of the cell / autolysis [1/2] {max 2}

(iii) Explain the roles of two named organelles in the formation of organelle Y. [3]

- Two named organelles: Rough endoplasmic reticulum [1/2] and Golgi apparatus [1/2]

At the rough ER

- Synthesis [1/2] and folding of hydrolytic enzymes found in lysosome [1/2]
- Transport of the hydrolytic enzymes to the Golgi apparatus via transport vesicles [1/2]

At the Golgi apparatus

- Chemically modifies the hydrolytic enzymes [1/2]
- Sort and package the hydrolytic enzymes into vesicles [1/2]
- Vesicles are released at the maturing face which then fuse to form lysosomes [1/2]

(iv) Compare the structural differences between organelle Y and a chloroplast.[2]

- Lysosome is bound by a single membrane [1/2] while chloroplast is bound by a double membrane [1/2]
- There is no extensive folding of membrane in lysosome [1/2] while the inner membrane of chloroplast is folded to form the thylakoid membrane [1/2]
- Lysosome contains hydrolytic enzymes [1/2] while chloroplast contains enzymes for Calvin cycle, electron carriers, circular DNA, ribosomes, etc. [1/2]
- Embedded on the thylakoid membrane are ATP synthase enzymes as well as a series of electron carriers that make up the electron transport chain for photophosphorylation, while the membrane of structure X does not contain embedded enzymes or electron carriers;;

[Total: 14]

2

(a) Use Table 2.1. and 2.2 to answer the following questions:

- (i) Explain how the change in DNA sequence of HbA β globin resulted in the changed amino acid sequence in HbS β globin and led to occurrence of sickle cell anaemia. [4]
- Missense mutation;
 - Single-base substitution of thymine by (replaced by) adenine of DNA sequence / gene;
 - Adenine(2nd base) replaced by uracil in 7th codon of mRNA;
 - Valine instead of glutamic acid in amino acid sequence;
 - Valine has hydrophobic R group vs glutamic acid has hydrophilic R group;
 - Decreases solubility of deoxygenated HbS;
 - When oxygen is low, HbS molecules precipitates out of solution to form rigid fibres;
 - Causes red blood cells to change from biconcave to sickle shape;
 - Lowered efficiency in oxygen transport $\rightarrow$ symptoms of sickle cell anaemia; [1/2m each, max 4]
- (ii) In another mutation in HbA, a single base substitution in the codon for the seventh amino acid occurred where the first G of the codon was replaced by U. Explain what effect this base substitution would have on β globin synthesis. [2]
- Nonsense mutation [1/2]
 - Base substitution results in the formation of a stop codon [1/2]
 - Translation of the beta-globin will terminate prematurely [1/2]
 - Produces a non-functional truncated protein [1/2]
- (b) While screening the blood of patients with sickle cell anaemia, the doctor found different nucleotide sequences in the chromosomal locus for haemoglobin. Given that the same Hb S haemoglobin is expressed, suggest **two** reasons for the differences noted. [2]

The differences in nucleotide sequences come from:

- Introns; which are spliced out and not expressed [1]
- Degenerate code/redundancy where each amino acid can be encoded by more than one codon of different nucleotide sequences. [1]

[Total :8]

3 (a) Explain the term “retroviruses”. [2]

- Retroviruses are viruses with RNA genome / RNA viruses [1/2]
- It carries the enzyme reverse transcriptase [1/2]
- When the retrovirus infects a cell, its RNA is used as a template to be reverse transcribed to DNA [1/2]
- The viral DNA is then integrated into the host's genome [1/2]
- The host cell's machinery then replicates and transcribes the viral DNA as if it were its own [1/2]

(b) Explain why enveloped viruses will not exit their host cells the same way as the T4 bacteriophage. [2]

- Enveloped viruses exit host cells through budding to obtain their viral envelop from the host cell [1/2]
- As well as viral glycoproteins which have been embedded into the host cell membrane [1/2]
- At specific locations in the host cell membrane known as exit points [1/2]
- T4 bacteriophages exit the host cells through cell lysis [1/2]
- In which the peptidoglycan cell wall of the bacterial host cell is hydrolysed by phage-coded lysozyme [1/2]
- If enveloped viruses exit in the same way as the T4 bacteriophage, the membrane and the glycoprotein embedded within would not become part of the viruses [1/2]

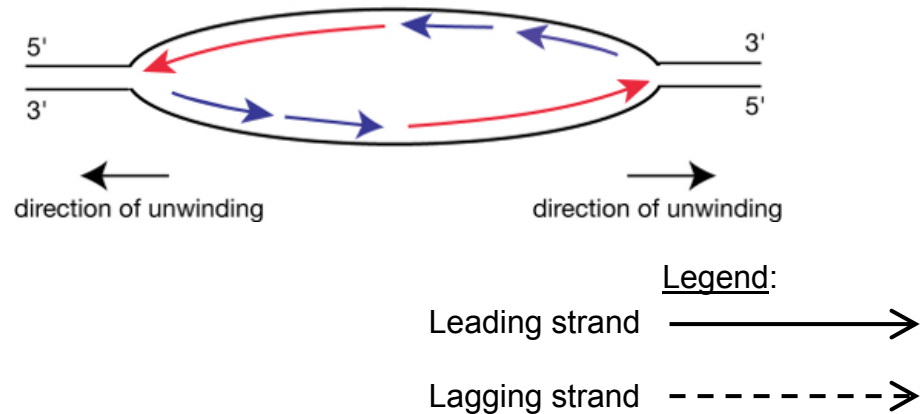
(c) (i) Describe the process of binary fission. [5]

- Circular DNA attaches to cell membrane / mesosome at its origin of replication [1/2]
- DNA helicase binds to origin of replication [1/2]
- And unwinds the double stranded DNA [1/2]
- Both parental DNA strands serve as templates for semi-conservative DNA replication [1/2]
- In which free DNA nucleotides bind to the parental strands via complementary base pairing [1/2]
- Primase catalyses the formation of RNA primers to provide free 3'OH ends [1/2]
- DNA polymerases catalyse the formation of phosphodiester bonds between adjacent DNA nucleotides [1/2]
- To synthesise the new DNA strands in the 5' to 3' direction with respect to the growing DNA strand [1/2]
- DNA replication occurs in a bidirectional manner until the entire DNA is replicated [1/2]
- Cytoplasmic division occurs as the cell elongates [1/2]
- And the membrane region between the two replicated DNA molecules grow toward each other to form a double layer across the cell [1/2]
- New cell wall layers are secreted within the membrane layers [1/2]
- This results in two genetically identical daughter cells [1/2]

(ii) Explain one difference between binary fission and the rolling circle mechanism of DNA replication that is significant to the initiation of DNA replication. [2]

	Binary fission	Rolling circle
Difference	None of the DNA strands is nicked / cut [1/2]	One DNA strand is nicked / cut [1/2]
Significance	RNA primers provide free 3'OH ends for DNA polymerase to add DNA nucleotides to [1/2]	Nicked strand provides the free 3'OH end for DNA polymerase [1/2]

- (d) The diagram shows part of a simplified but incomplete structure of replication bubble. Draw the two newly synthesized DNA strands using the legend provided. [2]



[Total: 13]

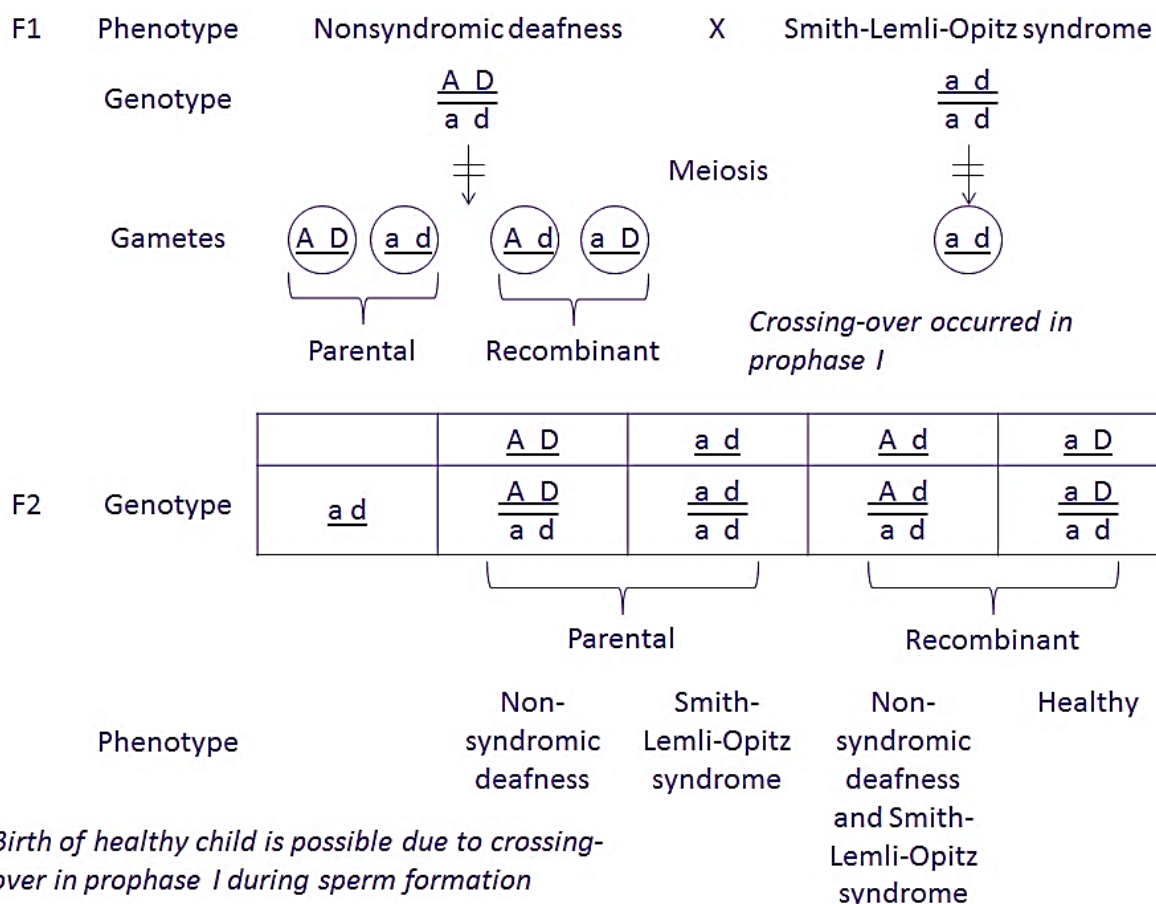
- 4 (a) (i) With reference to Fig 4.1, describe the process where the *p21* gene is transcribed into the *p21* mRNA in the presence of active p53 protein. [4]

- Active p53 serves as the activator and binds to the regulatory region (enhancer sequence) of *p21* gene [1/2]
- This causes a looping mechanism to bring the bound p53 close to the promoter of *p21* gene with the aid of DNA bending proteins [1/2]
- P53 then stabilises the transcription initiation complex which is formed at the promoter to upregulate transcription of *p21* gene [1/2]
- The transcription initiation complex is formed when general transcription factors recognise and bind to the TATA box in the promoter [1/2]
- And they then recruit RNA polymerase to bind to the promoter to form the transcription initiation complex [1/2]
- RNA polymerase unwinds the DNA double helix of the *p21* gene to expose the template strand [1/2]
- Free ribonucleotides bind to the template strand via complementary base pairing with the DNA nucleotides in the template strand [1/2]
- RNA polymerase moves along the template strand from 3' to 5' and catalyse the formation of phosphodiester bonds between adjacent ribonucleotides [1/2]
- Elongating the *p21* mRNA in the 5' to 3' direction with respect to the growing mRNA [1/2]

- Transcription of the *p21* gene continues until the RNA polymerase reaches the terminator sequence [1/2]
- (a) (ii) Describe how the mRNA of *E.coli* can give rise to different protein products, unlike that of the *p21* mRNA. [2]
- mRNA of *E.coli* (prokaryotic) is polycistronic [1/2]
 - It contains sequences coding for the amino acid sequences of multiple proteins punctuated with start and stop codons [1/2]
 - *p21* mRNA (eukaryotic) is monocistronic [1/2]
 - It contains sequences coding for the amino acid sequence of only *p21* with one set of start and stop codons [1/2]
- (b) State two other ways by which *p53* help prevent cancer. [2]
- *p53* acts as transcription factor to activate genes coding for DNA repair proteins [1/2]
 - To repair damages in DNA [1/2]
 - *p53* initiate apoptosis / programmed cell death [1/2]
 - If DNA damage is irreparable [1/2]
- (c) Describe 2 methods of post-translational control that can be employed in a cell to control the amount of active protein in a cell. [2]
- Proteolytic cleavage [1/2]
 - To convert inactive protein to its active form by removal of a short peptide sequence [1/2]
 - Biochemical modification e.g. phosphorylation [1/2]
 - To activate / inactivate proteins [1/2]
 - Ubiquitinylation [1/2]
 - Mark proteins for degradation by proteosomes [1/2]

[Total: 10]

- 5 (a)** A man suffering from nonsyndromic deafness but not Smith-Lemli-Opitz syndrome was homozygous for both *DHCR7* and *TECTα* genes. He married a woman suffering from Smith-Lemli-Opitz syndrome but not nonsyndromic deafness. They had a son who suffered only from nonsyndromic deafness. The son married another woman with only Smith-Lemli-Opitz syndrome, who gave birth to a healthy child without either disease.
- (i) Using the symbol **A/a** to represent the *TECTα* gene and **D/d** to represent the *DHCR7* gene, draw a genetic diagram to show how the son and his wife can give birth to a healthy child. [5]



(ii) Explain how you could use a statistical test to show that the inheritance of TECT α and DHCR7 genes does not follow Mendel's law. No calculation is needed. [2]

- Chi-square test [1/2]
- Calculated chi-square value is larger than the critical chi-square value at $p = 0.05$ and degree of freedom = 3 [1/2] OR p-value corresponding to calculated chi-squared is less than 0.05 at degree of freedom = 3 [1/2]
- This shows that there is significant differences between the expected Mendelian ratio of 1:1:1:1 and the observed results [1/2]
- The differences are not due to chance [1/2]

- (b) Apart from DHCR7, another enzyme required in the biosynthesis of cholesterol is squalene synthase, which is coded for by a gene found on chromosome 8. **Fig. 5.1** below shows a simplified biosynthetic pathway of cholesterol as well as vitamin D. Deficiency in vitamin D results in osteomalacia, a condition in which there is softening of bones.

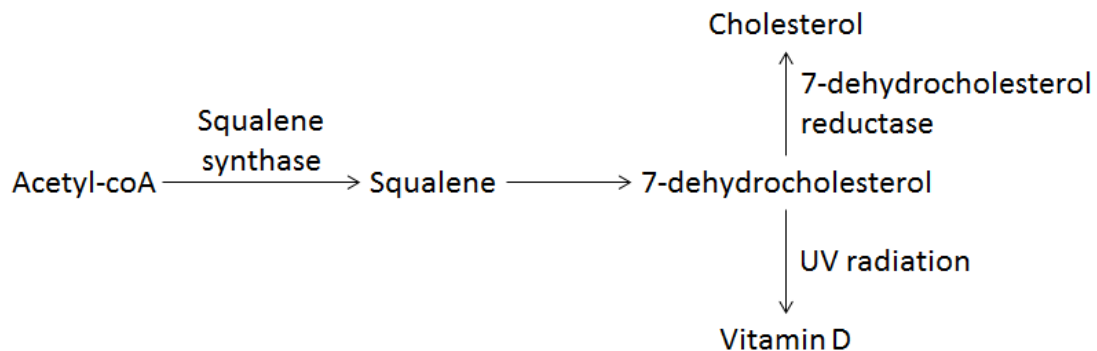


Fig. 5.1

A couple, both of them being heterozygous for both of the genes for DHCR7 and squalene synthase, exhibit a healthy phenotype with adequate level of cholesterol and vitamin D. They are expecting their first child in six months' time.

- (i) Explain the effect of independent assortment on their child's genotype. [3]

- Increase genetic variation / more possible genotypes [1/2]

During gamete formation

- Random orientation of the homologous pairs of chromosomes [1/2], 8 and 11 [1/2] at the equator during metaphase I [1/2]
- Subsequent separation of homologous chromosomes during anaphase I [1/2] would result in random assortment of the different alleles in each gene [1/2]
- Producing gametes with 2^n / different combinations of alleles in each parent [1/2]

- (ii) State the probability that their first child would suffer from both Smith-Lemli-Opitz syndrome and osteomalacia.

- $1/4$ [1]

[Total: 11]

6 (a) (i) Identify structure X. [1]

- Synaptic vesicle (containing neurotransmitters)

(iii) Explain the relationship between Neuron A and Neuron B during nervous transmission. [5]

- Neuron B is the presynaptic cell and Neuron A is the post-synaptic cell.
- When a depolarisation reaches Neuron B, voltage-gated Ca^{2+} channels open and Ca^{2+} enters Neuron B;
- This triggers synaptic vesicles in B to fuse with the pre-synaptic membrane of Neuron B and release their neurotransmitters by exocytosis;
- The neurotransmitters diffuse across the synaptic cleft and bind to receptors on the post-synaptic membrane of Neuron A;
- This triggers the opening of ligand-gated Na^{+} channels and results in an excitatory post-synaptic potential in Neuron A.

(b) Explain the role of ATP within Neuron B in synaptic transmission. [2]

- ATP is required for active transport of Ca^{2+} out of the presynaptic cell (after an Action potential) into the synaptic cleft;
- to maintain a low concentration of Ca^{2+} inside the cell ;
- so that during the next wave of depolarisation, Ca^{2+} can diffuse down their concentration gradient through the open voltage-gated Ca^{2+} channels on the the presynaptic cell membrane;
- ATP is required for the process of exocytosis where the influx of Ca^{2+} subsequently causes neurotransmitters to be released from the pre-synaptic cell;;
- ATP is required for the synthesis of acetylcholine in Neuron B;;

(c) Explain why transmission of impulse across a synapse is unidirectional.[2]

- Synaptic vesicles are only found in the axon terminal of the pre-synaptic cell::
- Receptors for neurotransmitter binding are only found in the post-synaptic cell membrane::

[Total:10]

7

a) (i) With reference to Figure 7 and theory of natural selection, explain the differences in the morphology of the llama camels in South America and the Camelus camels in Africa with the ancestral camel. [5]

- The llama are smaller in size while the Camelus have humps [1/2]
- due to genetic mutations [1/2] in the ancestral populations
- Reference to genetic variations in the ancestral populations [1/2]
- The environments of Africa and South America were different from North America / idea of different environmental conditions [1/2]

- → presence of selective pressures favouring the small size in the cold / mountainous region regions of S America [1/2] and the humps in the hot and dry / desert conditions of Africa. [1/2]
- Reference to natural selection [1/2]
- The better adapted individuals in their respective environments were able to survive and reproduce at a higher rate than those less adapted individuals [1/2].
- They pass their favorable alleles to the next generation [1/2].
- Over time, the gene pools and the camel populations in the two regions become significantly distinct from the ancestral camels and with each other. [1/2]
- Reference to the absence of gene flow between ancestral populations [1/2]
- Descent with modification of the ancestral characteristics / features [1/2]
- e.g. evolution of only two functional toes, supported by expanded pads for walking on snow or sand [1/2]

(b) Suggest why the South American camels have a longer evolutionary history than the African or Asian camels. [1]

- South America became connected to North America earlier than Asia / the ancestral camels would be able to migrate to South America earlier [1/2]
- Reference to South America linked to North America two million years ago compared to the formation of the land bridge across the Bering Strait one million years ago [1/2]

(c) Suggest why Arabian camels are found in Australia but they are no longer found in their place of origin in North America. [1]

- The geophysical forces (e.g. climate / ice-age / plate tectonics) resulted in conditions which were unfavorable to the ancestral camels → extinction / migration from North America
- Traders / humans brought the camels from North Africa and the Middle East to Australia by ship in the 1850s and the camels were able to adapt to the new environment

(d) The Arabian camels in Australia typically have sand coat color. However, there is a small percentage of camels which have albino coat. It is known that the albino coat is a recessive conditions and it reduces the life-span of the camel.

(i) Explain why population is considered the smallest unit of evolution. [2]

- Evolution is the sum of changes in the allele frequency in the gene pool of a population or species over time [1/2]
- A population is a group of individuals belonging to a particular species living in the same geographical region [1/2]
- An individual cannot evolve; natural selection acts on the individual [1/2] and it is the reproductive success of all these individuals in a population that will determine the proportion of different alleles in the next generation [1/2]
- Evolution by natural selection (the evolutionary force) can only be seen in the increasing proportion of sand coat colour camels in the population over time [1/2]

(ii) Explain whether it is likely that the albino camel will disappear completely from the population over time. Assume there is no introduction of new albino camels into the population. [3]

- Unlikely [1/2]
- Reference to diploidy where recessive allele is protected in the heterozygote condition (i.e. heterozygote protection) [1]
- In heterozygote (Gg) camel [1/2] the dominant allele will be expressed → the advantageous coat colour [1/2]
- When the heterozygote reproduce, the recessive allele (g) will also be passed to the next generation and remain in the gene pool [1/2]
- When two heterozygote interbreed, there is 25% chance of deriving a homozygote recessive camel (gg) showing the albino coat [1/2]

(iii) Explain how the evolutionary effect of genetic drift is affected by the sample size of the population. [2]

- Genetic drift refers to the **random changes in allele frequency** [1/2] that occurs in populations over generations
- It is **more pronounced in smaller population size**. [1/2]
- QV: Over 140 generations, allele frequency **fluctuates around 50%** for **population size of 2000** and **35-80%** for population **size 200**. [1/2]
- For population size 20, it rose from **50 to 75%** and then decreased to **0%** [1/2]

Section B

8 (a) Explain with examples, how environment may affect phenotype [8]

- The phenotype of an organism refers to the / observable characteristics of an individual (also accept: physical or chemical expression of the organism's genes) [1/2] while genotype refers to the genetic makeup of the organism or the alleles that an organism has [1/2]
- Genotype is the ultimate factor determining a phenotypic expression but in some cases [1/2], the environment affects the level of expression of the genes / affects the subsequent expression of the genetic potential [1/2].
- This is shown when genetically identical individuals develop differently in different environments. [1/2]
- Hence, the expression of a phenotype is affected by interaction of genotype and environmental factors. [1/2]

Named example 1: Effect of environmental conditions (e.g. light) on plant development / height (max 3 marks)

(Height in plants)

- The height of a plant is genetically-determined [1/2] (e.g. Mendel's tall variety of the garden peas plant) but growth depends on adequate light, water and soil conditions [1/2]

- A reduction in the supply of any one condition prevent the gene for height from exerting its full potential [1]

(Chlorophyll synthesis in plants)

- Although the ability to synthesize chlorophyll is genetically determined [1/2], light is a requirement [1/2]
- Evidence: seeds grown in the dark; such plants exhibit etiolation (e.g. stems are long and thin; seedlings are yellow) [1]

(Floral colours in *Hydrangea*)

- *Hydrangea* may have different floral colours despite carrying the same alleles; [1/2]
- The soil acidity, in which the plants grow affects the plants' ability to take up aluminium; [1/2]
- In acidic soils (pH 5.5 or lower), aluminum assumes a form that is easily absorbed by plant roots, and thus flowers are predominately blue; [1/2]
- In soils where the pH is 6.5 or higher / alkaline, aluminum is unavailable and flower color is pink purple; [1]
- Sometimes a single plant will have both blue and pink flowers because of varying soil conditions around the plant; [1]

(Height of yarrow plants)

- Height is genetically determined; [1/2]
- Cuttings from the same plant have the same genotype but grow differently at different altitudes / elevations; [1/2]
- Cuttings from one plant grew tall at lowest and highest elevation; [1/2]
- But remained short at mid-elevation; [1/2]

Named example 2: Effect of temperature on development of animal

(max 2 marks)

(Wing development in fruit-flies)

- The allele for vestigial wing in *Drosophila* / "fruit-flies" is recessive to that for long wing [1/2]
- However individuals which are homozygous for this allele [1/2] will only express the vestigial wings at low temperatures [1/2]
- Reference to vestigial wings at 21°C; [1/2] intermediate wings (26°C); long wings (31°C) [1/2]

Named example 3: Effect of diet on development of human / animal

(max 3 marks)

(Phenylketonuria in humans)

- Diet affects traits such as height, weight and intelligence in humans [1/2]
- Phenylalanine is metabolized by phenylalanine hydroxylase [1/2]
- Individuals with two copies of the mutant recessive alleles (homozygous recessive condition) do not have functional enzyme [1/2] → unable to break down the amino acid consumed through their diet [1/2] → phenylalanine accumulates in their bodies [1/2] → disease: phenylketonuria (PKU); mental retardation [1/2]
- Hence, these individuals need diet free from the particular amino acid [1/2]

(Reproductive system in honey bees)

- In a bee colony, the male bees or drones develop from unfertilized haploid eggs while the female bees develop from fertilized diploid eggs. [1/2]
- The worker bees are sterile while the queen bee is fertile
- Worker bees are smaller and have larger mouthparts and modified legs as compared to the queen bee; (they are phenotypically different even though genetically similar) ;
- The development of the female larvae to a queen bee or worker bee depends on the diet. [1/2]
- Once a particular female larva is selected to become the sexually mature queen bee, it is fed exclusively with royal jelly. [1/2]
- It is the high protein level in the royal jelly that stimulated the development of the female reproductive system. [1/2]
- Otherwise, it would be like the rest of the honey bee larvae which are fed royal jelly for the first few weeks after hatching (briefly) and then fed with a diet of honey and pollen. [1/2]

Named example 4: Effect of environment on development of human (max 3 marks)

(Pattern baldness in humans)

- male gender; premature pattern baldness due to an allele which is differentially expressed in the sexes; [1/2]
- both male homozygotes and heterozygotes develop bald patches; [1/2]
- only female homozygotes show balding; [1/2]
- expression of allele is probably triggered by testosterone; [1/2]
- females produce less of testosterone and thus seldom develop bald patches; [1/2]

(Skin colour in humans)

- exposure to the sun will result in the darkening of the skin / tanned skin;
- due to melanin production in cells;
- despite having an allele coding for fair skin;

Named example 5: Effect of environment on gene expression of E.coli cells (max 2 marks)

- The enzymes that metabolize lactose are coded by structural genes of the lac operon (e.g. β galactosidase, galactoside permease, galactoside transacetylase) [1/2]
- Synthesis of the enzymes depends on presence of the lactose in the environment of the prokaryote [1/2]; are synthesized when lactose is present which serves as an inducer for the activation of the lac operon [1/2]
- If nutrient medium / environment does not contain lactose or contains only glucose, the enzymes are not synthesized [1/2]

8 (b) Explain the effect of linkage on the phenotypic ratios from dihybrid crosses. [6]

- Linkage refers to genes/loci on the same chromosome;
- Alleles on the loci located close to each other tend to be inherited together;

- And do not undergo independent assortment/ do not follow Mendel's Law of Independent Assortment;
- Hence the (F1) offspring will display a predominance of parental types;
- And a small number of offspring with recombinant phenotypes/ of recombinants;
- Recombinants formed by crossing over occurring between the 2 loci (during prophase I);
- Where exchange of alleles between non-sister chromatids of a pair of homologous chromosomes;
- Crossing over is a chance event;
- (Relate distance to chance of crossing over to proportion of recombinants –)
 - o Proportion/Number of recombinants will depend on the distance between the 2 gene loci i.e. the closer the 2 gene loci on the chromosome, the less likely the chance of crossing over occurring between the 2 loci, hence the lower the proportion/number of recombinants (Accept the reverse argument);;
- If 2 genes are very tightly linked/ total linkage;
- They are inherited as a single unit;
- Hence only parental phenotypes are produced;
- Results of the dihybrid cross appears as a monohybrid cross;
- Eg. a cross between 2 double heterozygotes will produce a 3:1 phenotypic ratio in the offspring ;
- Instead of the expected 9:3:3:1 dihybrid ratio;
- Which is a result of independent assortment;
- Or a test cross will produce a 1:1 phenotypic ratio instead of the expected 1:1:1:1 dihybrid test cross ratio;

8(c) Explain how natural selection give rise to evolution. [6]

- In nature, populations tend to overproduce but population numbers remain constant;
- due to presence of certain environmental factors / selective agents;
- an example of selective agent such as predation/competition for limited food/ resources etc
- Natural selection is the process in which individuals that have certain heritable characteristics survive and reproduce at a higher rate than other individuals/ individuals have differential survival and reproduction;
- (Natural selection operates on the population in nature)
 - o individuals in a population show genetic variation (due to gene reshuffling during meiosis and mutations);
 - o the environmental factors act as a selective agent that selects for organism / individuals with the favourable combination of characteristics/traits;
 - o (these individuals will) survive longer and reproduce;

- o and the (favourable) alleles coding for these favourable traits are passed to/inherited by their offspring/ subsequent generations;
- o individuals with the unfavourable combination of characteristics/traits are selected against/ less likely to survive and reproduce;
- o and the alleles coding for these unfavourable traits are not passed to their offspring/ subsequent generations;
- o With time, the frequency of alleles that conferring an advantage increases; and
- o frequency of alleles that conferring a disadvantage decreases; (in the gene pool) of the population;
- The accumulated changes in time to the frequencies of many alleles in the gene pool of a population over time may result in evolutionary changes Or evolution can be measured by changes in the relative proportions of heritable variations in a population in successive generations;
- Accumulated changes in many alleles over a long period of time may result in new species being formed (speciation); OR
- Accumulation of different mutations plus genetic drift causes greater genetic differences between populations;
- Reproductive isolation is necessary for speciation to occur;
- Natural selection is more pronounced in small populations;

9 (a) Explain the principles of homeostasis with reference to regulation of blood glucose concentration by glucagon. [8]

Every (:) is ½ m

- *Definitions/Explanations of key terms or concepts* (mark under H)
 - Homeostasis is the maintenance of a constant internal environment;
 - Stating the norm/set point value for blood glucose level – blood glucose level is maintained at 80-120 mg/100 cm³ of blood;
 - Principle of self-regulating / self-adjusting mechanism;
 - Where the corrective mechanism is triggered by the entity which is to be regulated;
 - In regulation of blood glucose level it is by blood glucose itself;

(Principle of negative feedback is marked under N)

- *Receptor* (mark under R)
 - A receptor/detector detects any deviation from a set/reference point;
 - and responds by sending out signals/regulators to initiate corrective mechanism/ to effectors to correct the deviation;
 - When blood glucose level drops below the norm;
 - It is detected by α-cells of the Islets of Langerhans (in the pancreas);
 - which respond by secreting glucagon into the blood;
- *Effector* (mark under E)

- An effector receives the signals from the receptors; and
- carries out the corrective mechanism;
- Liver is the effector; *Reject*: muscles
- Any example of corrective mechanism eg. increase glycogenolysis which involves breaking down of glycogen to form glucose;
- Glucose is released into the blood stream;
- Blood glucose level raises (towards/back to the norm);
- *Negative Feedback* (mark under N)
 - Negative feedback as a disturbance in the system sets in motion a sequence of events which counteracts the disturbance;
 - and tends to restore the system to its original state; OR Negative feedback serves to bring about change in the opposite direction to restore stability;
 - Explanation of the higher blood glucose being detected by α -cells;
 - which respond by decreasing/stopping the release/secretion of glucagon; (Alternative answers for the 3rd and 4th bullets :
 - Excessive effects of glucagon is balanced by detection;
 - and negative feedback mechanism activated that inhibits the secretion of glucagon;)
 - Circulating glucagon destroyed in the liver;
 - therefore allowing the blood glucose levels to remain at the norm;

9 (b) Explain how species are formed with respect to physiological and behavioural isolation. [6]

Pre-requisites for speciation to occur (max 3 marks)

- Reproductive barriers [1/2] (e.g. behavioural isolation, physiological isolation) must be present to disrupt gene flow between 2 populations of the same species [1/2] (i.e. reproductive isolation between the 2 populations)
- As mutations occur independently in each population; [1/2]
- And each population undergoes natural selection [1/2] and genetic drift
- Over time: accumulation of genetic differences / 2 gene pools diverge [1/2]
- Eventually, (fertile / sexually mature) members of one population cannot interbreed with members of the other population to form viable and fertile offspring → species formed / reproductive isolation is complete [1/2]

Physiological isolation (max 3 marks for examples)

Examples of physiological differences as pre-zygotic barriers

- Mechanical isolation [1/2] - the mating/ copulation apparatus of mature adults from one population are not compatible with that of the other population [1/2] → unsuccessful mating [1/2]
- Gametic isolation [1/2] – male gametes cannot survive in the reproductive tract of the female of the other population [1/2] → unable to fertilise the egg [1/2]
- Temporal isolation [1/2] - e.g. Different mating / sexual maturation seasons [1/2]

Examples of physiological differences as post-zygotic barriers

- Non-disjunction occurring during meiosis resulting in polyploidy/ an extra set of chromosomes [1/2]
- Autopolyploid individuals (often plants) are reproductively isolated from diploid individuals of the original species [1/2]
- Or allopolyploid individuals formed from the mating between individuals from 2 different species are reproductively isolated from both parent species [1/2]
- but may be able to reproduce among themselves, eventually forming a new species by themselves [1/2]

Behavioural isolation

- 2 populations may develop over time significant differences such as differences in dancing rituals, calls or other sexual displays to attract potential mates [1/2]
→ prevents individuals from the 2 different populations from being attracted to each other / recognize each other as potential mates [1/2] → contributing to reproductive isolation of the 2 populations
- This is a form of pre-zygotic barrier [1/2] while physiological isolation may exist as pre-zygotic or post-zygotic barriers

Max 5 marks if behavioural isolation not discussed

(9c) Compare between a ribosome and a centriole. [6]

Similarities:

- Both are non-membrane bound organelles [1/2]
- Both are composed of proteins [1/2]
- Both can be found in animal cells [1/2]

Differences:

Features	Ribosome	Centriole
Composition	• Consists of rRNA and ribosomal proteins [1/2] • Made up of large and small subunits [1/2]	• Consists of tubulin protein [1/2] • Made up of 9 triplets of microtubules [1/2]
Numbers in cell	• Numerous [1/2]	• One pair of rod-like structure arranged perpendicular to each other [1/2]
Location in cell	• Bound to rough endoplasmic reticulum or found freely in the cytoplasm (also found in mitochondria and chloroplast) [1/2]	• Found next to the nucleus at the centrosome [1/2]
Type of cells found in	• Both animal and plant cells (also in prokaryotes) [1/2]	• Only animal cells [1/2]
Function	• Protein synthesis [1/2]	• Act as the microtubule-organising centre involved in spindle fibre formation [1/2]